INTRODUCTION TO FINANCIAL ACCOUNTING

ACCT 220

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Chapter 10

- Reporting and Interpreting Bonds

  - What are bonds?
  - Recording interest expense and reporting bonds on the balance sheet

Debt versus Equity

- Corporations can raise money through
  - Equity
  - Debt

- What should a corporation consider when raising capital?
  - Debt has the following benefits and costs:

- Pros
  - Current stockholders maintain control
  - Interest is tax deductible (dividends are not)

- Cons
  - Higher risk of bankruptcy
    - No obligation to repay shareholders at a certain time
    - BUT do need to repay creditors regardless of profit or loss

What are bonds?

- Definition: <u>publically</u> traded debt of a firm

Bonds

- Critical Terminology
  - Principal: amount payable at the maturity of the bond and on which the periodic cash interest payments are computed
    - Also called: par value, face amount, face value
  - Coupon Rate: rate that determines the cash interest payment per period as stated on the bond contract
    - Also called: stated rate, contract rate

Bonds

- **Potential Bond Features**

Unsecured Bond (AKA Debenture)	Secured Bond
No assets are pledged as a guarantee of repayment at maturity	Specific assets are pledged as a guarantee of repayment at maturity
(No backup for creditor)	(Creditor will seize assets, for example foreclose on a property)

Bonds

- **Potential Bond Features**

  - **Callable**: may be repaid early by the issuer (typically the issuer must pay a premium)
    - Also called the early retirement of debt

  - **Convertible**: bond maybe converted into common stock of the issuer

  - **Zero Coupon**: no interest payments are made over time (interest is rolled into repayment of principal at maturity date)

What a bond used to look like

Credit Ratings

- Credit rating agencies assign default ratings
  - Agencies include
    - Moody’s
    - S&P
    - Fitch
  - Default rating = likelihood of bankruptcy (inability to repay debts)

What is a junk bond?

- If rating is below investment grade ➔ junk bond
  - Why does this matter?
    - The rating affects whether or not certain institutional investors can buy or hold a company's debt

Reporting Bond Transactions

- The bond itself will provide the following:
  - Principal (AKA par value, face value)
  - Cash interest payment (based on coupon rate)

- BUT also need to know market interest rate (AKA yield or effective interest rate)
  - This is determined by what the market is willing to pay for the bond

Reporting Bond Transactions

- There are two interest rates?!
  1. Coupon Rate: set by issuer
  2. Market Rate: determined by the market

- And things get complicated whenever the coupon rate ≠ market rate

Companies generally try to have a coupon rate similar to market rate, but cannot always figure out exactly what market rate is ahead of time

Reporting Bond Transactions

- Implications of the two interest rates:

Coupon Rate		Market Rate		Sold at a	Explanation
Coupon	>	Market	➔	Premium	-Coupon payments more than cover interest payments expected by the market -Buyers willing to pay more than face value
Coupon	=	Market	➔	Par	-Market demanded interest same as coupon rate
Coupon	<	Market	➔	Discount	-Coupon payments do NOT cover interest payments expected by the market -Buyers NOT willing to pay face value

Reporting Bond Transactions: Sold at Par

- Easiest because no discount or premium to calculate
- For each period:

  Interest expense = coupon rate * principal

Reporting Bond Transactions: Sold at Par

- When sell bond to investors:

Cash (A+)	X	
Bond Payable (L+)		X

*X=face value

- When interest payments are made:

Interest Expense (SE-)	Y	
Cash (A-)		Y

*Y=interest rate * principal

Reporting Bond Transactions: Sold at Par

- Numerical Example:
  - A company sells a $100,000 2 year bond with 10% coupon rate. Cash interest payments are made annually. The market rate is also 10%
  - Sell Bond:

Cash (A+)	100,000	
Bond Payable (L+)		100,000

  - Interest after 1 year:

Interest Expense (SE-)	10,000	
Cash (A-)		10,000

  - Interest after 2 years:

Interest Expense (SE-)	10,000	
Cash (A-)		10,000

  - Repay:

Bond Payable (L-)	100,000	
Cash (A-)		100,000

When Bonds are NOT Sold at Par

- When the coupon rate ≠ market rate, bond will not sell for face value
  - Instead, the market will price the bond (meaning will pay cash for) based on the interaction of coupon and market rates
    - Discount (coupon rate<market rate) -> Company receive less than face value
    - Premium (coupon rate>market rate) -> Company receive more than face value

When Bonds are NOT Sold at Par

- How does the market price bonds?
  - Investors willing to pay:

    Present value of principal

    + Present value of interest payments

    Market price of bond

    Interest payments based on coupon rate

    Interest rate for PV calculation = market interest rate

Reporting Bond Transactions: Sold at a Discount

- Investors are not willing to pay full face value for a bond if the coupon rate is less than the market rate.
  - Will pay less than face value and the company will amortize the difference (i.e. discount) over time

Reporting Bond Transactions: Sold at a Discount

- When sell the bond to investors
- Journal Entry

Cash (A+)	Y	
Discount on Bond Payable (XL+)	(Z-Y)	
Bond Payable (L+)		Z

Z = face value of bond

Y = cash received from investors = market price= PV of principal and interest payments using market rate

Reporting Bond Transactions: Sold at a Discount

- Need to amortize this discount using the effective interest method
  - Journal Entry

Interest Expense (SE-)	A
Discount on Bond Payable (XL-)	(A-B)
Cash (A-)	B

A= (Bond Payable – Discount on Bond) x Market Rate

B= Coupon Payment = Bond Payable x Coupon Rate

Notice: interest expense changes each period because discount is reduced over time

Reporting Bond Transactions: Sold at a Discount

To figure out the interest expense and adjustments to the discount account, create a schedule:

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate * Net Book Value)	(Interest Expense – Interest Payment)	(Face Value – Discount)

Reporting Bond Transactions: Sold at a Discount

Cash given to bond holder according to coupon rate

Interest expense as reporting on the income statement

Reduction in Discount -> Increased NBV

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate * Net Book Value)	(Interest Expense – Interest Payment)	(Face Value – Discount)

Reporting Bond Transactions: Sold at a Discount

Alternative Way of Thinking: allocates the discount created at issuance over time

Cash given to bond holder according to coupon rate

Interest expense as reporting on the income statement

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate * Net Book Value)	(Interest Expense – Interest Payment)	(Face Value – Discount)

Reporting Bond Transactions: Sold at a Discount

Numerical example: Company sells a $10,000 face value, 3 year bond with a 5% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 10%. The company's fiscal year end is 12/31.

Using excel, the bond sells for

Present Value of the Principal (single amount):	PV(.1,3,0,-10000)	=$7,513.15
+Present Value of Interest (annuity):	PV(.1,3,-500)	=$1,243.43
What Investors are Willing to Pay (also, Net Book Value)		=$8,756.57

Discount=Face Value – PV of cash flows = $10,000 – $8,756.57 = $1,243.43

Reporting Bond Transactions: Sold at a Discount

Numerical example: Company sells a $10,000 face value, 3 year bond with a 5% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 10%. The company's fiscal year end is 12/31.

Discount=Face Value – PV of cash flows = $10,000 – $8,756.57 = $1,243.43

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate*Net Book Value)	(Interest Expense – Interest Payment)	(Face Value – Discount)
1/1/20				8,756.57
12/31/20	(.05*10,000)=500			
12/31/21	(.05*10,000)=500			
12/31/21	(.05*10,000)=500			

Reporting Bond Transactions: Sold at a Discount

Numerical example: Company sells a $10,000 face value, 3 year bond with a 5% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 10%. The company's fiscal year end is 12/31.

Discount=Face Value – PV of cash flows = $10,000 – $8,756.57 = $1,243.43

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate*Net Book Value)	(Interest Expense – Interest Payment)	(Face Value – Discount)
1/1/20				8,756.57
12/31/20	(.05*10,000)=500	(.1*8,756.57)=875.66	(875.66-500)= 375.66	(8,756.57+375.66)= 9,132.23
12/31/21	(.05*10,000)=500			
12/31/22	(.05*10,000)=500			

Reporting Bond Transactions: Sold at a Discount

Numerical example: Company sells a $10,000 face value, 3 year bond with a 5% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 10%. The company's fiscal year end is 12/31.

Discount=Face Value – PV of cash flows = $10,000 – $8,756.57 = $1,243.43

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Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate*Net Book Value)	(Interest Expense – Interest Payment)	(Face Value – Discount)
1/1/20				8,756.57
12/31/20	(.05*10,000)=500	(.1*8,756.57)=875.66	(875.66-500)= 375.66	(8,756.57+375.66)= 9,132.23
12/31/21	(.05*10,000)=500	(.1*9,123,23)=913.22	(913.22-500)=413.22	(9,123.23+413.22)= 9,545.45
12/31/22	(.05*10,000)=500			

Reporting Bond Transactions: Sold at a Discount

Numerical example: Company sells a $10,000 face value, 3 year bond with a 5% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 10%. The company's fiscal year end is 12/31.

Discount=Face Value – PV of cash flows = $10,000 – $8,756.57 = $1,243.43

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Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate*Net Book Value)	(Interest Expense – Interest Payment)	(Face Value – Discount)
1/1/20				8,756.57
12/31/20	(.05*10,000)=500	(.1*8,756.57)=875.66	(875.66-500)= 375.66	(8,756.57+375.66)= 9,132.23
12/31/21	(.05*10,000)=500	(.1*9,132.23)=913.22	(913.22-500)=413.22	(9,123.23+413.22)= 9,545.45
12/31/22	(.05*10,000)=500	(.1*9,545.45)=954.54	(954.45-500)=454.54	(9,545.45+454.54)= 9,999.99*

*Rounding errors regularly happen

In Class Practice Problem 1

Reporting Bond Transactions: Sold at a Premium

- Investors are willing to pay more than the face value for a bond if the coupon rate is greater than the market rate.
  - Will get more than face value and amortize the difference (the premium) over time

Reporting Bond Transactions: Sold at a Premium

- When sell the bond to investors:
- Journal Entry

Cash (A+)	Y	
Premium on Bond Payable (L+)		(Y-Z)
Bond Payable (L+)		Z

Z= face value of bond

Y = cash received from investors

Reporting Bond Transactions: Sold at a Premium

- Need to amortize this premium using the effective interest method
  - Journal Entry

Interest Expense (SE-)	A	
Premium on Bond Payable (L-)	(B-A)	
Cash (A-)		B

A= (Bond Payable + Premium on Bond) x Market Rate

B= Coupon Payment = Bond Payable x Coupon Rate

Notice: interest expense changes each period because premium is reduced over time

Reporting Bond Transactions: Sold at a Premium

To figure out the interest expense and adjustments to the premium account, create a schedule

Notice different from discount

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate * Net Book Value)	(Interest Payment-Interest Expense)	(Face Value + Premium)

Reporting Bond Transactions: Sold at a Premium

Cash given to bond holder according to coupon rate

Interest expense as reporting on the income statement

Reduction in premium

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate * Net Book Value)	(Interest Payment-Interest Expense)	(Face Value + Premium)

Reporting Bond Transactions: Sold at a Premium

Cash given to bond holder according to coupon rate

Interest expense as reporting on the income statement

Alternative Way of Thinking: allocates the premium created at issuance over time

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate * Net Book Value)	(Interest Payment-Interest Expense)	(Face Value + Premium)

Reporting Bond Transactions: Sold at a Premium

Numerical example: Company sells a $10,000 face value, 3 year bond with a 10% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 5%. The company's fiscal year end is 12/31.

Using excel, the bond sells for PV(.05,3,0,10000) +PV(.05,3,1000) = $11,361.62

Present Value of the Principal (single amount):	PV(.05,3,0,-10000)	=$8.638.38
+Present Value of Interest (annuity):	PV(.05,3,-1000)	=$2,723.25
What Investors are Willing to Pay (also, Net Book Value)		= $11,361.62

Premium=Cash from Investors- Face Value = $11,361.62 - $10,000 = $1,361.62

Reporting Bond Transactions: Sold at a Premium

Numerical example: Company sells a $10,000 face value, 3 year bond with a 10% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 5%. The company's fiscal year end is 12/31.

Premium=PV of cash flows - Face Value = $11,361.62 - $10,000 = $1,361.62
Interest Payment = Coupon Rate * Face Value = 10% * $10,000=$1,000

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate*Net Book Value)	(Interest Payment-Interest Expense)	(Face Value + Premium)
1/1/20				11,361.62
12/31/20	(.1*10,000)=1,000			
12/31/21	(.1*10,000)=1,000			
12/31/22	(.1*10,000)=1,000			

Reporting Bond Transactions: Sold at a Premium

Numerical example: Company sells a $10,000 face value, 3 year bond with a 10% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 5%. The company's fiscal year end is 12/31.

Premium=PV of cash flows - Face Value = $11,361.62 - $10,000 = $1,361.62

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate*Net Book Value)	(Interest Payment - Interest Expense)	(Face Value + Premium)
1/1/20				11,361.62
12/31/20	(.1*10,000)=1,000	(.05*11,361.62)=568.08	(1,000-568.08)=431.92	(11,361.62-431.92)=10,929.71
12/31/21				
12/31/22				

Reporting Bond Transactions: Sold at a Premium

Numerical example: Company sells a $10,000 face value, 3 year bond with a 10% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 5%. The company's fiscal year end is 12/31.

Premium=PV of cash flows - Face Value = $11,361.62 - $10,000 = $1,361.62

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate*Net Book Value)	(Interest Payment - Interest Expense)	(Face Value + Premium)
1/1/20				11,361.62
12/31/20	(.1*10,000)=1,000	(.05*11,361.62)=568.08	(1,000-568.08)=431.92	(11,361.62-431.92)=10,929.71
12/31/21	(.1*10,000)=1,000	(.05*10,929.70)=546.49	(1,000-546.49)=453.51	(10,929.70-453.51)= 10,476.19
12/31/22				

Reporting Bond Transactions: Sold at a Premium

Numerical example: Company sells a $10,000 face value, 3 year bond with a 10% coupon rate on 1/1/2020. Interest is paid annually. The market rate is 5%. The company's fiscal year end is 12/31.

Premium=PV of cash flows - Face Value = $11,361.62 - $10,000 = $1,361.62

	Amortization Schedule			
Date	Interest Payment	Interest Expense	Amortization	Net Book Value
	(Coupon Rate * Principal)	(Market Rate*Net Book Value)	(Interest Payment - Interest Expense)	(Face Value + Premium)
1/1/20				11,361.62
12/31/20	(.1*10,000)=1,000	(.05*11,361.62)=568.08	(1,000-568.08)=431.92	(11,361.62-431.92)=10,929.71
12/31/21	(.1*10,000)=1,000	(.05*10,929.70)=546.49	(1,000-546.49)=453.51	(10,929.70-453.51)= 10,476.19
12/31/22	(.1*10,000)=1,000	(.05*10,476.19)=523.81	(1,000-523.81)=476.19	(10,476.19-475.19)=10,000.00

In Class Practice Problem 2

Takeaways

- What are bonds?
- Recording interest expense and reporting bonds on the balance sheet